# SMART SHOPPING TROLLEY

**A PROJECT REPORT**

*Submitted by*


| | |
|---|---|
| **GOGUL R** | **830122106006** |
| **MOHAMMED ABBAS M** | **830122106021** |
| **NARENDHIRAN M** | **830122106023** |
| **BARANI S** | **830122106301** |


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**TIRUCHIRAPPALLI-620012**


**ANNA UNIVERSITY: CHENNAI 600 025**

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# BONAFIDE CERTIFICATE

Certified that this project report **"SMART SHOPPING TROLLEY"** is the bonafide work of **"GOGUL R (830122106006), MOHAMMED ABBAS M (830122106021), NARENDHIRAN M (830122106023)** and **BARANI S (830122106301)"** who carried out the project under my supervision.

**SIGNATURE**

**Dr.M.ELANGOVAN M.E., Ph.D.,**
**HEAD OF THE DEPARTMENT**

Department of ECE
Government College of
Engineering Srirangam
Trichy-12

**SIGNATURE**

**Dr.D.KALAIARASAN M.E., Ph.D.,**
**SUPERVISOR**

Department of ECE
Government College of
Engineering Srirangam
Trichy-12

Submitted on the project work examination held on…….... at Government College of Engineering Srirangam, Tiruchirappalli - 620012.

**INTERNAL EXAMINER** **EXTERNAL EXAMINER**

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# ABSTRACT

In this project an advanced automated solution designed to enhance the efficiency and convenience of the retail shopping experience. This project integrates an ESP32 microcontroller with RFID technology, a load cell-based weight verification system, a 16×2 LCD display, and wireless communication to provide a real-time, user-friendly billing mechanism. In this system, each product is equipped with an RFID tag, and an RFID-RC522 reader is used to scan the items placed in the trolley. Upon scanning, the ESP32 retrieves the product information and updates the total cost, which is displayed on the LCD screen. To ensure accuracy and prevent fraudulent activities, a load cell interfaced through an HX711 module continuously monitors the weight of the items and compares it with predefined values stored in the system. Any mismatch triggers an alert through a buzzer. The system is powered by a rechargeable 18650 Li-ion battery, making it portable and suitable for real-time usage in supermarkets. Additionally, the built-in Wi-Fi capability of the ESP32 enables communication with a local server or mobile application, allowing real-time monitoring of shopping data and facilitating digital payment integration. This smart trolley eliminates the need for traditional checkout queues, reduces billing time, minimizes human error, and improves overall customer satisfaction. The proposed system is cost-effective, scalable, and can be widely implemented in modern retail environments to support automated and contactless shopping.

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# CHAPTER 1

# INTRODUCTION

In today's fast-paced world, the retail industry is continuously evolving to improve customer convenience and operational efficiency. One of the major challenges faced by supermarkets and shopping malls is the long waiting time at billing counters, especially during peak hours. Traditional billing systems require manual scanning of products at the checkout point, which leads to delays, increased workload for staff, and reduced customer satisfaction.

To overcome these limitations, the Smart Shopping Trolley System has been proposed as an innovative solution that automates the billing process within the shopping cart itself. This system utilizes modern embedded technology by integrating an ESP32 microcontroller with RFID-based product identification, load cell-based weight verification, and a real-time display unit. Each product is equipped with an RFID tag, allowing customers to scan items as they place them into the trolley. The system instantly retrieves product details and updates the total cost, which is displayed on a 16×2 LCD screen.

In addition to automated billing, the inclusion of a load cell ensures product verification by measuring the weight of items placed in the trolley. This feature helps in preventing mismatches and enhances the reliability of the system. The ESP32's built-in Wi-Fi capability enables seamless communication with a local server or mobile application, supporting real-time monitoring and future integration of digital payment methods such as QR-based transactions. The entire system is powered by a rechargeable Li-ion battery, making it portable and suitable for practical deployment in supermarkets.

By eliminating the need for traditional checkout queues, this smart trolley significantly reduces billing time, minimizes human errors, and improves the overall shopping experience. Thus, the Smart Shopping Trolley represents a step toward smart retail automation, combining embedded systems, wireless communication, and sensor technology to create an efficient and user-friendly solution for modern shopping environments.

# CHAPTER 2

# LITERATURE SURVEY

The concept of a smart shopping trolley has gained significant attention in recent years due to the increasing demand for automation in the retail sector. Various researchers and developers have proposed different techniques to improve the efficiency of billing systems and enhance the customer shopping experience. **R. Kaur et al. (2016)** developed a smart shopping cart using barcode technology to simplify the billing process. Their system reduced manual effort but still required scanning at the checkout counter, which did not completely eliminate waiting time. **S. Patil and P. Joshi (2017)** introduced an RFID-based smart trolley system where products are automatically detected using RFID tags. This method improved scanning efficiency and reduced billing time. However, their system lacked a proper mechanism for verifying product authenticity. **M. Sharma et al. (2018)** proposed an embedded system using a microcontroller to calculate the total bill in real time and display it on an LCD screen. Their design improved automation but did not include any anti-theft or verification features.

**A. Verma and R. Singh (2019)** enhanced the smart trolley concept by integrating wireless communication technologies such as Wi-Fi for real-time data transmission. This allowed the billing data to be monitored on a central server, making the system more scalable and efficient. **K. N. Rao et al. (2020)** introduced a load cell-based verification system along with RFID technology. Their system compared the expected weight of products with the actual weight in the trolley, thereby improving accuracy and preventing fraudulent activities. **P. Kumar and S. Mehta (2021)** proposed an IoT-based smart shopping system that included mobile application integration and digital payment features. This

system provided a complete smart retail solution but increased overall system complexity and cost.

From the above studies, it is observed that while RFID technology significantly improves billing efficiency, the addition of weight verification and wireless communication enhances system reliability and functionality. The proposed Smart Shopping Trolley system combines these features using an ESP32 microcontroller, RFID reader, load cell, LCD display, and Wi-Fi connectivity to create a cost-effective and efficient solution.

# CHAPTER 3

# SYSTEM ANALYSIS

## 3.1 PROBLEM STATEMENT

In modern retail environments such as supermarkets and shopping malls, customers frequently experience long waiting times at billing counters, especially during peak hours and weekends. The traditional billing system relies on manual scanning of each product at the checkout counter, which significantly increases the time required for billing. This results in customer dissatisfaction and inefficient utilization of time. Moreover, the conventional system lacks real-time billing capabilities, as customers are unaware of the total cost of items until they reach the billing counter. This can lead to inconvenience, particularly when customers exceed their budget unknowingly.

Another major issue is the absence of an effective verification mechanism to prevent fraudulent activities such as product misplacement or incorrect billing. Human errors during scanning and billing are also common, which further affects the accuracy and reliability of the system. Therefore, there is a need for an automated, efficient, and reliable system that can perform real-time billing, reduce queue time, and ensure accurate product verification. The Smart Shopping Trolley system is designed to address these challenges by integrating advanced embedded technologies.

## 3.2 EXISTING SYSTEM

The existing billing system in most retail stores is based on barcode scanning technology. In this system, customers collect items while shopping and proceed to a centralized billing counter, where a cashier scans each product using a barcode scanner. The system then calculates the total bill and generates a receipt. Although barcode-based systems have improved billing accuracy

compared to manual entry methods, they still depend heavily on human intervention. The entire billing process is carried out only at the checkout counter, which leads to congestion and long queues.

Additionally, barcode scanning requires direct line-of-sight between the scanner and the barcode, which can sometimes slow down the process. The system also does not provide any mechanism for customers to track their spending in real time while shopping. Furthermore, the existing system does not include any form of weight verification or anti-theft mechanism, making it vulnerable to errors and potential misuse. As a result, while the current system offers basic functionality, it lacks efficiency, automation, and security.

### 3.3 DISADVANTAGES OF EXISTING SYSTEM

The traditional shopping and billing system has several limitations that affect both customers and store management. Firstly, the most significant disadvantage is the time-consuming billing process. Customers must wait in long queues at the billing counter, which leads to frustration and reduced customer satisfaction. Secondly, the system is prone to human errors. Mistakes in scanning, incorrect billing, or missing items can occur due to manual handling, leading to financial discrepancies.

Another major drawback is the lack of real-time cost tracking. Customers are unable to view the total amount of their purchases while shopping, which can result in overspending. In addition, the absence of a proper verification mechanism makes the system vulnerable to theft or incorrect item placement. There is no method to ensure that the items scanned match the items placed in the trolley. Moreover, the dependency on centralized billing counters increases operational workload and requires additional staff, thereby increasing operational costs. Thus, the existing system is inefficient, time-consuming, and lacks the necessary features required for modern smart retail environments.

## 3.4 PROPOSED SYSTEM

To overcome the limitations of the existing system, the Smart Shopping Trolley system is proposed as an advanced and automated solution. This system integrates multiple technologies such as RFID, load cell-based weight verification, ESP32 microcontroller, LCD display, and Wi-Fi communication. In the proposed system, each product is equipped with an RFID tag, and an RFID reader is installed on the shopping trolley. When a customer places an item in the trolley, the RFID reader automatically scans the product and retrieves its information. The ESP32 microcontroller processes this data and updates the total cost in real time.

A 16×2 LCD display is used to show the product details and the cumulative bill amount, allowing customers to monitor their expenses while shopping. This eliminates the need to wait until the checkout stage for billing information. To enhance system reliability, a load cell is integrated with the trolley to measure the weight of the items. The measured weight is compared with the expected weight of scanned products. If a mismatch is detected, an alert is generated using a buzzer, thereby preventing incorrect billing or theft.

The ESP32's built-in Wi-Fi capability allows the system to transmit data to a server or mobile application, enabling real-time monitoring and future integration of digital payment systems such as QR code-based payments. The entire system is powered by a rechargeable battery, making it portable and suitable for practical implementation. This proposed system significantly improves efficiency, accuracy, and user experience in retail shopping.

## 3.5 ADVANTAGES OF PROPOSED SYSTEM

Smart Shopping Trolley system offers several advantages over the traditional billing system. One of the primary advantages is the reduction in billing time. Since items are scanned automatically as they are placed in the trolley, there is no need for a separate billing process at the checkout counter. This eliminates long queues and saves time. Another important advantage is real-time billing. Customers can view the total cost of their purchases instantly on the LCD display, which helps them manage their budget effectively.

The system also improves accuracy by minimizing human intervention. Automated scanning and processing reduce the chances of errors in billing. The integration of a load cell provides an additional layer of security by verifying the weight of items. This helps in preventing fraud and ensures that only correctly scanned items are included in the bill. Furthermore, the use of Wi-Fi connectivity enables real-time data monitoring and supports future enhancements such as digital payment integration and inventory management.

The system is portable, cost-effective, and easy to use, making it suitable for deployment in modern retail environments. Overall, the proposed Smart Shopping Trolley system enhances efficiency, accuracy, security, and customer satisfaction, making it a valuable solution for smart retail automation.

# CHAPTER 4

# HARDWARE COMPONENTS

## 4.1 EMBEDDED SYSTEMS

A microprocessor-based computer hardware and software system called an embedded system is created to carry out a particular task, either independently or as a component of a larger system. An integrated circuit built to do computing for real-time processes is at the heart of the system.

From a single microcontroller to a group of linked processors with networks and peripherals, complexity can range from having no user interface to having intricate graphical user interfaces. Depending on the job for which it is created, an embedded system's complexity varies greatly.

Applications for embedded systems include hybrid cars, avionics, digital watches, microwaves, and more. Embedded systems consume up to 98 percent of all produced microprocessors.

### 4.1.1 Overview of embedded systems

An embedded system is a special-purpose computer system designed to perform one or a few dedicated functions, often with real-time computing constraints. It is usually embedded as part of a complete device including hardware and mechanical parts. In contrast, a general-purpose computer, such as a personal computer, can do many different tasks depending on programming. Embedded systems have become very important today as they control many of the common devices we use.

Since the embedded system is dedicated to specific tasks, design engineers can optimize it, reducing the size and cost of the product, or

increasing the reliability and performance. Some embedded systems are mass-produced, benefiting from economies of scale.

Physically, embedded systems range from portable devices such as digital watches and MP3 players, to large stationary installations like traffic lights, factory controllers, or the systems controlling nuclear power plants. Complexity varies from low, with a single microcontroller chip, to very high with multiple units, peripherals and networks mounted inside a large chassis or enclosure.

In general, "embedded system" is not an exactly defined term, as many systems have some element of programmability. For example, Handheld computers share some elements with embedded systems — such as the operating systems and microprocessors which power them — but are not truly embedded systems, because they allow different applications to be loaded and peripherals to be connected.

Embedded systems provide several functions:

- Monitor the environment; embedded systems read data from input sensors. This data is then processed and the results displayed in some format to a user or users.
- Control the environment; embedded systems generate and transmit commands for actuators.
- Transform the information; embedded systems transform the data collected in some meaningful way, such as data compression/decompression.

Although interaction with the external world via sensors and actuators is an important aspect of embedded systems, these systems also provide functionality specific to their applications. Embedded systems typically execute applications such as control laws, finite state machines, and signal processing algorithms.

These systems must also detect and react to faults in both the internal computing environment as well as the surrounding electromechanical systems.

There are many categories of embedded systems, from communication devices to home appliances to control systems. Examples include:

• Communication devices

Eg: modems, cellular phones

• Home Appliances

Eg: CD player, VCR, microwave oven

• Control Systems

Eg: Automobile anti-lock braking systems, robotics, satellite control

### 4.1.2 Block diagram of an Embedded systems

An embedded system usually contains an embedded processor. Many appliances that have a digital interface -- microwaves, VCRs, cars -- utilize embedded systems. Some embedded systems include an operating system. Others are very specialized resulting in the entire logic being implemented as a single program. These systems are embedded into some device for some specific purpose other than to provide general purpose computing. A typical embedded system is shown in Fig

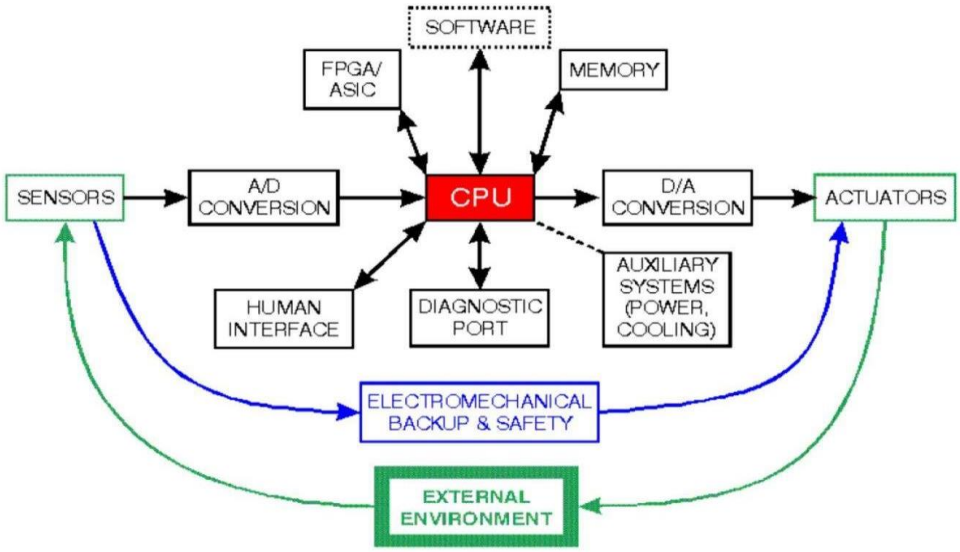



**Fig 4.1 Block diagram of a typical embedded system**

## 4.2 MICROCONTROLLER

A microcontroller is a compact integrated circuit designed to govern a specific operation in an Embedded System. A typical microcontroller includes a processor, memory, and input/output(I/O) peripherals on a single chip. Sometimes referred to as an embedded controller or microcontroller unit (MCU), microcontrollers are found in vehicles, robots, office machines, medical devices, mobile radio transceivers, vending machines, and home appliances, among other devices. They are essentially simple miniature personal computers (PCs) designed to control small features of a larger component, without a complex front-end operating system (OS).

### 4.2.1 Core elements of a Microcontroller

- The processor (CPU) -- A processor can be thought of as the brain of the device. It processes and responds to various instructions that direct the microcontroller's function.

This involves performing basic arithmetic, logic, and I/O operations. It also performs data transfer operations, which communicate commands to other components in the larger embedded system.

● Memory -- A microcontroller's memory is used to store the data that the processor receives and uses to respond to instructions that it's been programmed to carry out. A microcontroller has two main memory types:

1. Program memory, which stores long-term information about the instructions that the CPU carries out. Program memory is non-volatile memory, meaning it holds information over time without needing a power source.

2. Data memory, which is required for temporary data storage while the instructions are being executed. Data memory is volatile, meaning the data it holds is temporary and is only maintained if the device is connected to a power source.

● I/O peripherals -- The input and output devices are the interface for the processor to the outside world. The input ports receive information and send it to the processor in the form of binary data. The processor receives that data and sends the necessary instructions to output devices that execute tasks external to the microcontroller.

### 4.2.2 Supporting elements of a Microcontroller

● Analog to Digital Converter (ADC) -- An ADC is a circuit that converts analog signals to digital signals. It allows the processor at the center of the microcontroller to interface with external analog devices, such as sensors.

● Digital to Analog Converter (DAC) -- A DAC performs the inverse function of an ADC and allows the processor at the center of the microcontroller to communicate its outgoing signals to external analog components.

● System bus -- The system bus is the connective wire that links all components of the microcontroller together.

● Serial port -- The serial port is one example of an I/O port that allows the microcontroller to connect to external components. It has a similar function to a USB or a parallel port but differs in the way it exchanges bits.

## 4.3 HARDWARE REQUIRED

- ESP32 Microcontroller
- RFID Reader (RC522)
- RFID Tags
- Load Cell
- HX711 Load Cell Amplifier
- 16×2 LCD Display (I2C)
- Buzzer
- Power Supply (Li-ion Battery)

## 4.4 ESP32 MICROCONTROLLER

The ESP32 microcontroller is the central processing unit of the system. It is responsible for controlling all the connected modules and executing the program logic. The ESP32 is chosen due to its high performance, low power consumption, and built-in Wi-Fi capability. It processes data received from the RFID reader, calculates the total bill, verifies weight using the load cell, and updates the display accordingly. It also enables wireless communication with a server or mobile application.

### 4.4.1 ESP32 Block Diagram

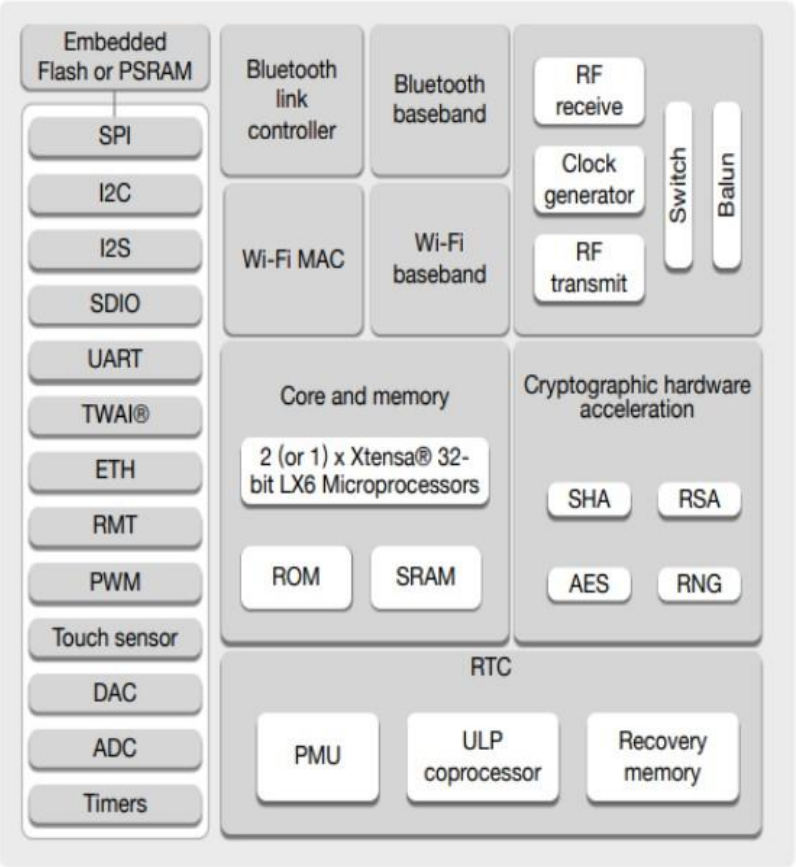



**Fig 4.2 Block diagram of ESP32**

### 4.4.2 Overview of ESP32 Architecture

### 1. Core and Memory

- The ESP32 contains 2 (or 1) Xtensa® 32-bit LX6 microprocessors, which act as the brain of the system.
- It supports dual-core processing, enabling multitasking and improved performance.
- Memory components include:
  - ROM – Stores firmware boot code
  - SRAM – Used for runtime data and program executive

## 2. Communication Interfaces

The left side of the block diagram shows multiple communication peripherals:

- SPI (Serial Peripheral Interface) – High-speed communication with sensors and displays
- I2C (Inter-Integrated Circuit) – Used for connecting low-speed devices like RTCs and EEPROMs
- I2S (Inter-IC Sound) – Audio data communication
- SDIO – Interface for SD cards
- UART – Serial communication for debugging and data transfer
- TWAI (CAN Bus) – Used in automotive and industrial systems
- ETH (Ethernet) – Wired network communication
- RMT (Remote Control Module) – Infrared and signal transmission
- PWM (Pulse Width Modulation) – Motor control and LED dimming

## 3. Wireless Communication Modules

- Wi-Fi MAC & Wi-Fi Baseband
    - Handles wireless networking and data packet processing
- Bluetooth Link Controller & Baseband
    - Supports Bluetooth Classic and BLE (Bluetooth Low Energy)
- These modules enable seamless wireless connectivity for IoT devices.

**4. RF (Radio Frequency) Section**

- Includes:
  - RF Transmit & Receive modules
  - Clock Generator
  - Switch and Balun
- Responsible for converting digital signals into radio waves and vice versa.
- Ensures stable wireless communication with minimal interference.

**5. Cryptographic Hardware Acceleration**

ESP32 includes dedicated hardware for security:

- AES (Advanced Encryption Standard) – Data encryption
- SHA (Secure Hash Algorithm) – Data integrity
- RSA – Secure communication
- RNG (Random Number Generator) – Security key generation

**6. RTC (Real-Time Clock) Subsystem**

The RTC block operates independently and supports low-power operations:

- PMU (Power Management Unit) – Controls power consumption
- ULP Coprocessor – Handles tasks in sleep mode
- Recovery Memory – Retains data during low-power states

### 7. Analog and Sensor Features

- ADC (Analog-to-Digital Converter) – Reads analog signals
- DAC (Digital-to-Analog Converter) – Outputs analog signals
- Touch Sensors – Capacitive touch input detection
- Timers – Used for scheduling and delays

## 4.4.3 ESP32 Pin Description

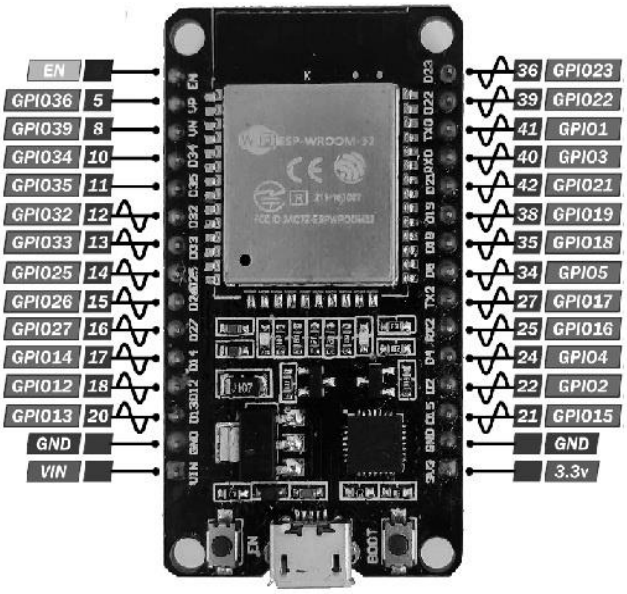



**Fig 4.3 Pin description of ESP32**

## 4.4.4 Overview of Pin Configuration in ESP32

### 1. Power Pins

- VIN: Input voltage (typically 5V from USB or external source)
- 3.3V: Regulated output voltage
- GND: Ground reference

**2. GPIO (General Purpose Input/Output) Pins**

- ESP32 has multiple GPIO pins (GPIO0–GPIO39)
- Can be configured as input or output
- Some pins are input-only (e.g., GPIO34–GPIO39)
- Used for LEDs, switches, relays, etc.

**3. Analog Pins**

ADC (Analog-to-Digital Converter)

- Converts analog signals into digital values
- ADC channels are mapped to multiple GPIOs (e.g., GPIO32–GPIO39)

DAC (Digital-to-Analog Converter)

- GPIO25 → DAC1
- GPIO26 → DAC2
- Used for generating analog output signals

**4. Communication Interfaces**

1. SPI (Serial Peripheral Interface)
   - Pins: MOSI, MISO, SCK, SS
   - Used for high-speed devices like displays and SD cards
2. I2C (Inter-Integrated Circuit)

   Default Pins:

   - SDA → GPIO21
   - SCL → GPIO22

3. UART (Universal Asynchronous Receiver/Transmitter)
    - TX → GPIO1
    - RX → GPIO3
    - Used for serial communication and debugging

**5. PWM (Pulse Width Modulation)**

- Almost all GPIO pins support PWM
- Used for:
    - Motor speed control
    - LED brightness control

**6. Touch Sensor Pins**

ESP32 has built-in capacitive touch sensors:

- Touch pins include GPIO0, GPIO2, GPIO4, GPIO12–GPIO15, GPIO27, GPIO32, GPIO33
- Used in touch-based interfaces like buttons and sliders

**7. RTC GPIO Pins**

- RTC (Real-Time Clock) pins work during deep sleep mode
- Useful for low-power applications
- Examples: GPIO0, GPIO2, GPIO4, GPIO12–GPIO15

**8. Special Function Pins**

- EN (Enable): Resets the board when pulled LOW
- GPIO0: Used for boot mode selection

- GPIO2: Must be LOW or floating during boot in some cases

**9. Input-Only Pins**

- GPIO34, GPIO35, GPIO36, GPIO39
- These pins cannot be used as outputs
- Commonly used for sensors

## 4.5 RFID READER (RC522)

The RFID RC522 module is used for identifying products placed in the trolley. Each product is attached with an RFID tag containing a unique identification number. When a product is brought near the RFID reader, it detects the tag and sends the data to the ESP32. This allows automatic scanning of items without manual intervention. The RFID reader operates using SPI communication and works at 3.3V.s

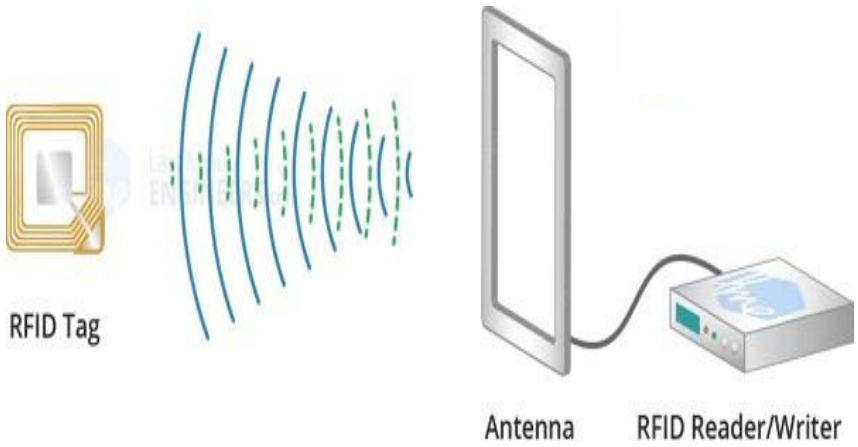



**Fig 4.4 RFID Technology**

RFID or Radio Frequency Identification system consists of two main components, a transponder/tag attached to an object to be identified, and a Transceiver also known as interrogator/Reader.

The RC522 RFID Reader module is designed to create a 13.56MHz electromagnetic field that it uses to communicate with the RFID tags (ISO 14443A standard tags). The reader can communicate with a microcontroller over a 4-pin Serial Peripheral Interface (SPI) with a maximum data rate of 10Mbps. It also supports communication over I2C and UART protocols.

### 4.5.1 Overview of RFID Reader/Writer Module

The RC522 RFID Reader module is designed to create a 13.56MHz electromagnetic field that it uses to communicate with the RFID tags (ISO 14443A standard tags). The reader can communicate with a microcontroller over a 4-pin Serial Peripheral Interface (SPI) with a maximum data rate of 10Mbps. It also supports communication over I2C and UART protocols.

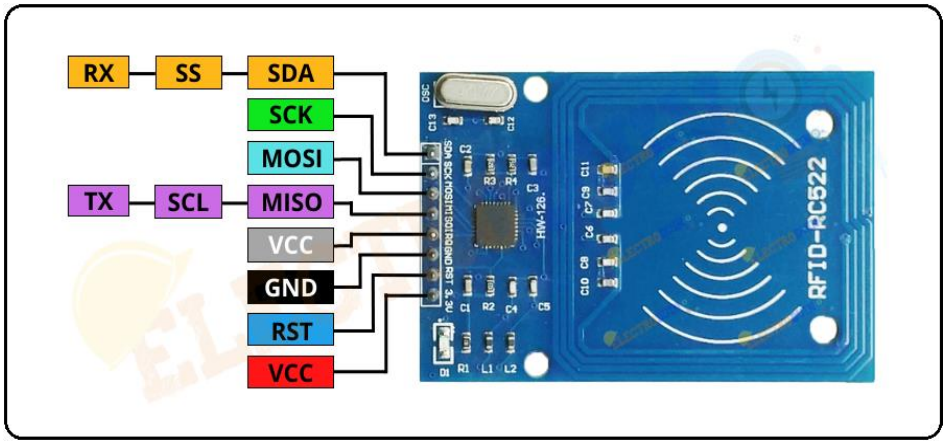



**Fig 4.5 RFID Pin Configuration**

- VCC: supplies power for the module. This can be anywhere from 2.5 to 3.3 volts. You can connect it to 3.3V output from your Arduino. Remember connecting it to 5V pin will likely destroy your module!
- RST: is an input for Reset and power-down. When this pin goes low, hard power-down is enabled. This turns off all internal current sinks including the oscillator and the input pins are disconnected from the outside world. On the rising edge, the module is reset.

- GND: is the Ground Pin and needs to be connected to GND pin on the Arduino.
- IRQ: is an interrupt pin that can alert the microcontroller when RFID tag comes into its vicinity.
- MISO / SCL / Tx pin: acts as Master-In-Slave-Out when SPI interface is enabled, acts as serial clock when I2C interface is enabled and acts as serial data output when UART interface is enabled.
- MOSI (Master Out Slave In): is SPI input to the RC522 module.
- SCK (Serial Clock): accepts clock pulses provided by the SPI bus Master i.e. Arduino.
- SS / SDA / Rx pin : acts as Signal input when SPI interface is enabled, acts as serial data when I2C interface is enabled and acts as serial data input when UART interface is enabled. This pin is usually marked by encasing the pin in a square so it can be used as a reference for identifying the other pins.

| Frequency Band | Common Frequency | Communication range | Data Rate |
|---|---|---|---|
| Low Frequency | 125KHz | 15cm-20cm | Low |
| High Frequency | 13.56MHz | 10cm-1m | Low to Moderate |
| Ultra-high Frequency | 868-928MHz | 3m-7m | Moderate to High |
| Microwave | 2.45 & 5.0 GHz | 10m-15m | High |

**Table 4.1 RFID Frequency Bands**

## 4.6 RFID TAGS

RFID tags are a type of tracking system that uses smart barcodes in order to identify items. RFID is short for “radio frequency identification”, and as such, RFID tags utilize radio frequency technology. These radio waves transmit data from the tag to a reader, which then transmits the information to an RFID computer program. RFID tags are frequently used for merchandise, but they can also be used to track vehicles, pets, and even patients with Alzheimer’s disease. An RFID tag may also be called an RFID chip. RFID tag includes microchip with radio antenna mounted on substrate which carries 12 Byte unique Identification number.

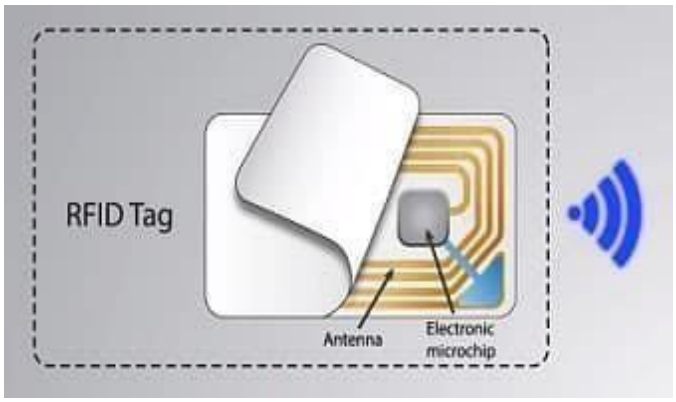



**Fig 4.6 RFID Tags**

### 4.6.1 Components of a RFID System

- A RFID tag: It consists of a silicon microchip attached to a small antenna and mounted on a substrate and encapsulated in different materials like plastic or glass veil and with an adhesive on the back side to be attached to objects.
- A reader: It consists of a scanner with antennas to transmit and receive signals and is responsible for communication with the tag and receives the information from the tag.

- A Processor or a Controller: It can be a host computer with a Microprocessor or a microcontroller which receives the reader input and process the data.

### 4.6.2 Types of RFID Tags

- **Passive Tags** – It is the cheaper version using no battery. The Tag uses radio energy transmitting from the reader. So, the Reader must be close to the tag to transfer energy to power the Tag. Since the tags have unique serial number, the reader can recognize them individually.
- **Active Tags**– These have an on-board battery and periodically transmits ID signals to the reader.
- **Battery Assisted Passive or BAP**– These Tags have small battery on board and will be activated in the presence of signals from the reader.
- **Read only Tags** – These have a unique factory assigned serial number used as the key for the data base.
- **Read/ Write Tags** – These can write object specific data give by the system user.
- **Field programmable Tags**– These can write once but read many times. Black tags can be written with an electronic product code by the user.

There are two main types of RFID tags: battery-operated and passive. As the name suggests, battery-operated RFID tags contain an onboard battery as a power supply, whereas a passive RFID tag does not, instead working by using electromagnetic energy transmitted from an RFID reader.

Battery-operated RFID tags might also be called active RFID tags. Passive RFID tags are a much more economical choice than active RFID tags, and cost around 20 cents each. This makes them a popular choice for supply chain management, race tracking, file management, and access control applications.

While a passive RFID tag does not require a direct line of sight to the RFID reader, it has a much shorter read range than an active RFID tag. They are small in size, lightweight, and can potentially last a lifetime.

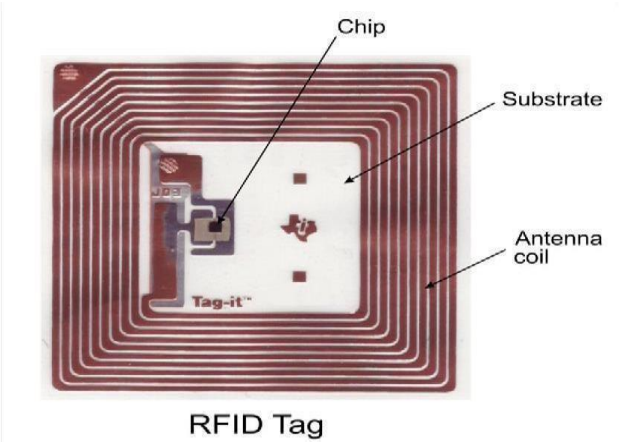



**Fig 4.7 RFID Tag Inside**

## 4.7 LOAD CELL

The load cell is used to measure the weight of items placed in the trolley. It converts the applied force (weight) into an electrical signal. This component plays a crucial role in verifying whether the scanned product matches the actual item placed in the trolley. It helps in detecting errors or fraudulent activities.

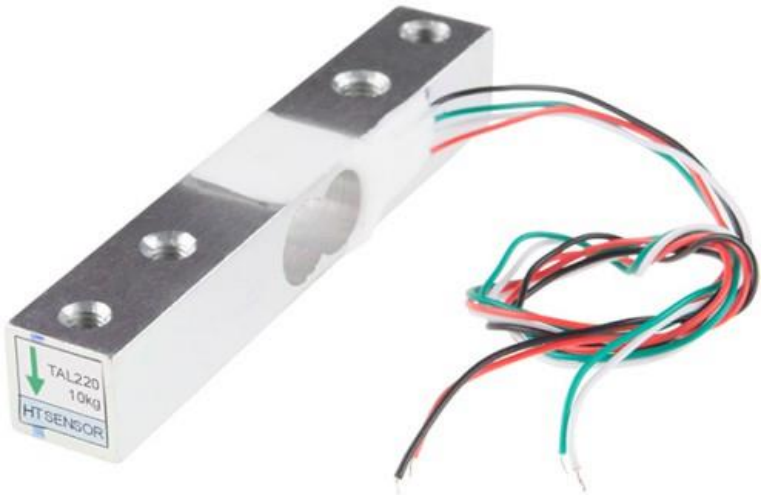



**Fig 4.8 Load Cell (Straight Bar)**

A load cell is a physical element (or transducer if you want to be technical) that can translate pressure (force) into an electrical signal. So, what does that mean? There are three main ways a load cell can translate an applied force into measurable reading.

### 4.7.1 Hydraulic Load Cells

Hydraulic load cells use a conventional piston and cylinder arrangement to convey a change in pressure by the movement of the piston and a diaphragm arrangement which produces a change in the pressure on a Bourdon tube connected with the load cells.

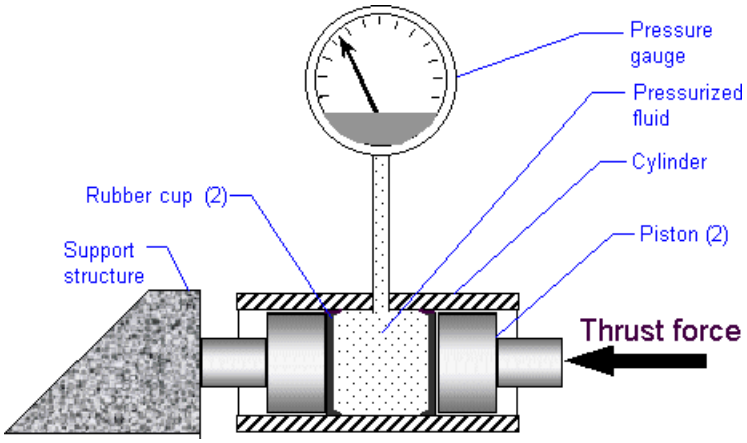



**Fig 4.9 Hydraulic Load Cells**

### 4.7.2 Strain Gauge Load Cells

And lastly (though there are many other less common load cell set ups), there is a strain gauge load cell, which is a mechanical element of which the force is being sensed by the deformation of a (or several) strain gauge(s) on the element.

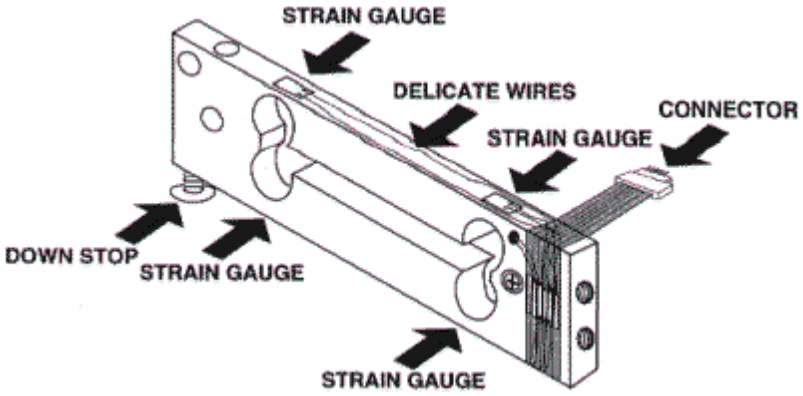



**Fig 4.10 Strain Gauge Load Cells**

In bar strain gauge load cells, the cell is set up in a "Z" formation so that torque is applied to the bar and the four strain gauges on the cell will measure the bending distortion, two measuring compression and two tension. When these four strain gauges are set up in a wheatstone bridge formation, it is easy to accurately measure the small changes in resistance from the strain gauges.

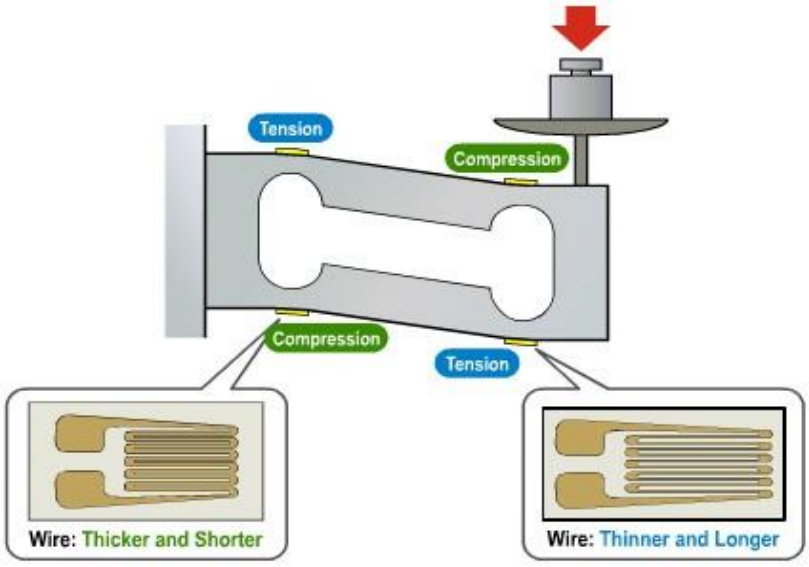



**Fig 4.11 Strain Gauge Load Cells Z formation**

### 4.7.3 Amplifiers and Wheatstone Bridge

This is where an amplifier, such as the HX711 or the NAU7802 comes in handy. A good way of taking small changes in resistance and turning it into something more measurable is using a wheatstone bridge. A wheatstone bridge is a configuration of four resistors with a known voltage applied like this:

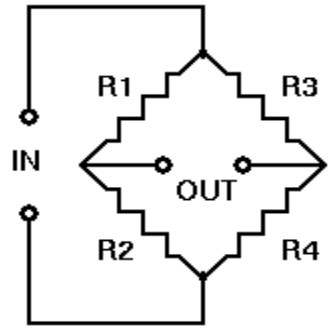



**Fig 4.12 Wheatstone Bridge**

where Vin is a known constant voltage, and the resulting Vout is measured. If the next is true

$$R1/R2 = R3/R4$$

then Vout is 0, but if there is a change to the value of one of the resistors, Vout will have a resulting change that can be measured and is governed by the following equation using Ohm's law:

$$Vout = [(R3/(R3 + R4)) - R2/(R1 + R2)]Vin$$

By replacing one of the resistors in a wheatstone bridge with a strain gauge, we can easily measure the change in Vout and use that to assess the force applied.

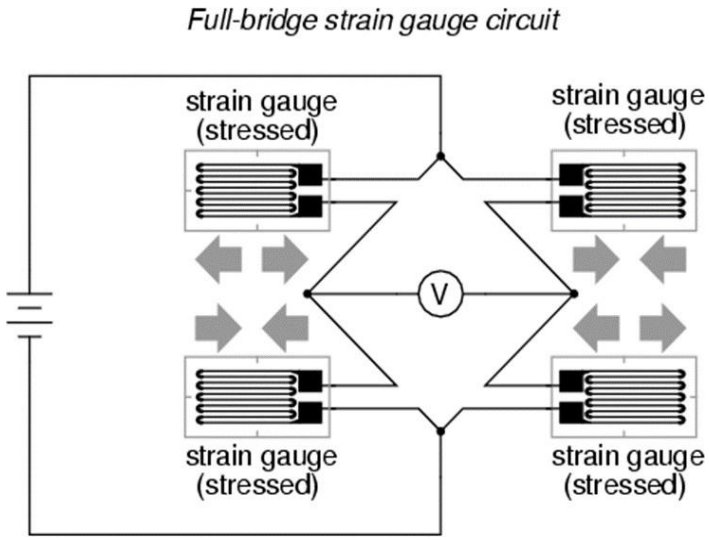



**Fig 4.13 Full-Bridge strain gauge circuit**

## 4.8 HX711 LOAD CELL AMPLIFIER

The HX711 module is used to amplify and convert the small electrical signal from the load cell into a readable digital signal. It provides high accuracy and is specially designed for weight measurement applications. The module communicates with the ESP32 using digital pins and ensures precise weight readings.

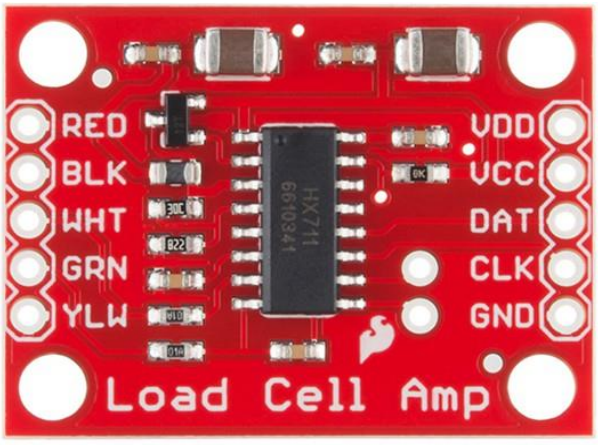



**Fig 4.14 HX711 Integrated Circuit**

Based on Avia Semiconductor's patented technology, HX711 is a precision 24-bit analog-to-digital converter (ADC) designed for weigh scales and industrial control applications to interface directly with a bridge sensor.

Load cells use a four-wire Wheatstone bridge configuration to connect to the HX711. These are commonly colored RED, BLK, WHT, GRN and YLW. Each color corresponds to the conventional color coding of load cells.

- Red (Excitation+ or VCC)
- Black (Excitation- or GND)
- White (Amplifier-, Signal- or Output-)
- Green (A+, S+ or O+)
- Yellow (Shield)

The YLW pin acts as an optional input that is not hooked up to the strain gauge but is utilized to ground and shield against outside EMI (electromagnetic interference).

### 4.8.1 HX711 Block Diagram

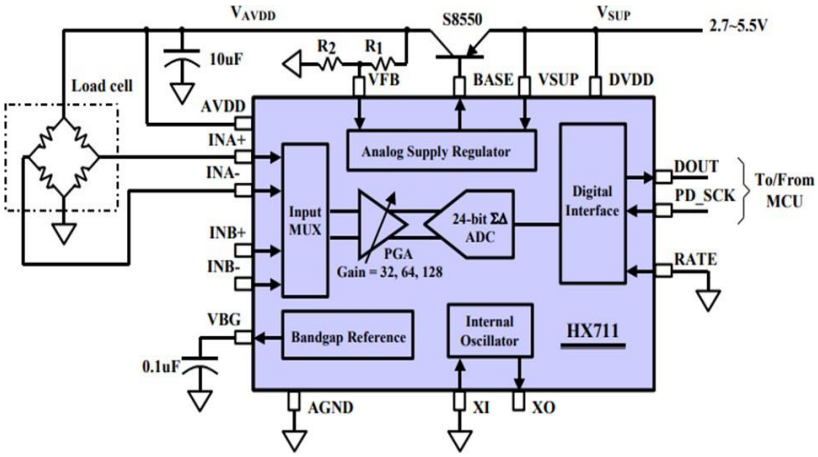



**Fig 4.15 HX711 Block Diagram**

### 4.8.2 HX711 Pin Description

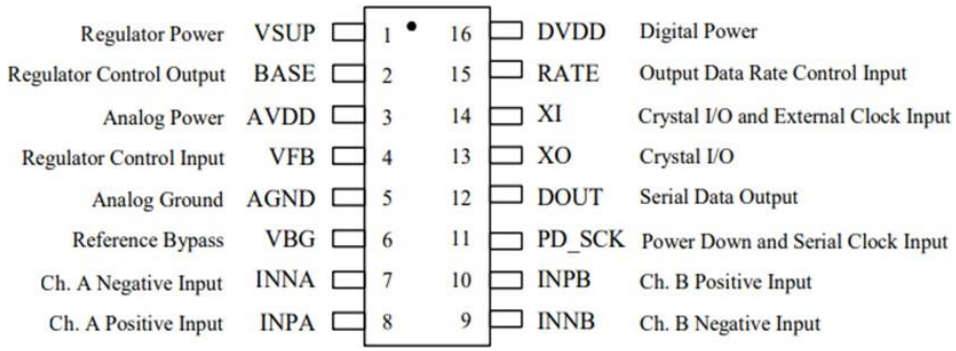



**Fig4.16 HX711 Pins**

### 4.9 16×2 LCD DISPLAY (I2C)

The 16×2 LCD display is used to show product details and the total bill amount to the user. The I2C interface reduces the number of connections required, using only two communication lines (SDA and SCL). The display provides real-time feedback to the customer, improving usability and transparency.

A liquid crystal display (LCD) is a thin, flat display device made up of any number of color or monochrome pixels arrayed in front of a light source or reflector. It uses very small amounts of electric power, and is therefore suitable for use in battery powered electronic devices.

| Pin no | Name | Function |
|---|---|---|
| 1 | GND | Ground 0V |
| 2 | VCC | Supply voltage;5V(4.7V-5.3V) |
| 3 | VEE | Contrast Adjustment; through a variable resistor |

| | | |
|---|---|---|
| 4 | RS | Register selects; Selects command when low and data when high |
| 5 | R/W | Low to write to the register, High to read from the register |
| 6 | EN | Sends data to data pins when a high to low pulse is given |
| 7 | DB0 | Data bus bit 0 |
| 8 | DB1 | Data bus bit 1 |
| 9 | DB2 | Data bus bit 2 |
| 10 | DB3 | Data bus bit 3 |
| 11 | DB4 | Data bus bit 4 |
| 12 | DB5 | Data bus bit 5 |
| 13 | DB6 | Data bus bit 6 |
| 14 | DB7 | Data bus bit 7 |
| 15 | LED+ | Backlight VCC(5V) |
| 16 | LED- | Backlight Ground(0V) |

**Table 4.2 Pin Description of LCD**

The PCF8574 IIC/I2C module for the LCD16x2 is an interface module that is used to communicate with the LCD16x2 display using the I2C (Inter-Integrated Circuit) protocol. Usually, the LCD16x2 requires a minimum of six GPIO pins (if we use it in 4-Bit mode), i.e. higher numbers of GPIOs get engaged.



**Fig 4.17 Interfacing 16x2 LCD Using I2C**

The I2C LCD Driver Module Only requires 2 GPIOs for an LCD16x2 display, which will be helpful as we need to utilize a smaller number of GPIOS.It simplifies the wiring and control of the LCD by reducing the number of pins required for communication.This module is compatible with both LCD16x2 displays as well as LCD20x4 display.

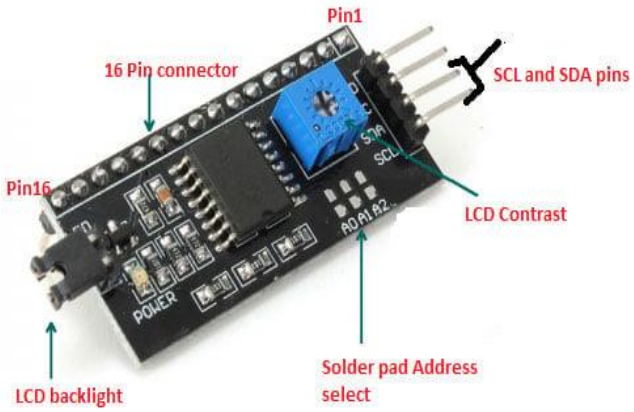



**Fig 4.18 I2C LCD Module**

This LCD Module includes the POT to adjust the contrast of the LCD and pull-up resistors for SDA and SCL lines. So, we don't need any additional connections. We just plugin the I2C LCD Module at the back of the 16×2 LCD and make connections between the microcontroller and the I2C LCD Module.

| LCD I2C Pin | ESP8266 Pin |
|---|---|
| GND | GND |
| VCC | VCC |
| SDA | D2 |
| SCL | D1 |

**Table 4.3 Connection setup of ESP32, I2C Module and LCD**

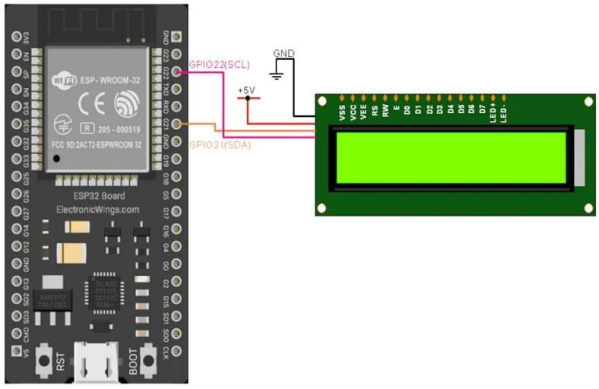



**Fig 4.19 Interfacing 16x2 LCD with ESP32 Using I2C**

## 4.10 BUZZER

The buzzer is used as an alert mechanism in the system.It is activated when there is a mismatch between the expected weight and the measured weight, indicating a possible error or unauthorized action. It provides an audible alert to notify the user immediately.

A buzzer is a mechanical, electromechanical, magnetic, electromagnetic, electro-acoustic or piezoelectric audio signaling device. A piezo electric buzzer can be driven by an oscillating electronic circuit or other audio signal source.

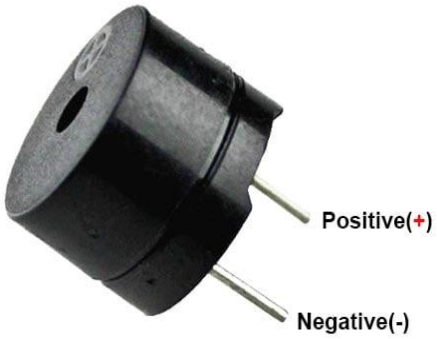



**Fig 4.20 Buzzer**

Piezo buzzer is an electronic device commonly used to produce sound. Light weight, simple construction and low price make it usable in various applications like car/truck reversing indicator, computers, call bells etc. Piezo

buzzer is based on the inverse principle of piezo electricity discovered in 1880 by Jacques and Pierre Curie.

It is the phenomena of generating electricity when mechanical pressure is applied to certain materials and the vice versa is also true. Such materials are called piezo electric materials. Piezo electric materials are either naturally available or manmade. Piezoceramic is class of manmade material, which poses piezo electric effect and is widely used to make disc, the heart of piezo buzzer. When subjected to an alternating electric field they stretch or compress, in accordance with the frequency of the signal thereby producing sound.

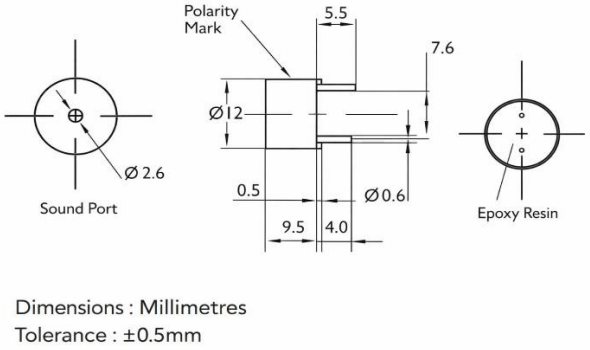



**Fig 4.21 Dimensions of Buzzer**

### 4.10.1 Dimensions and Specifications

All measurements are given in millimeters (mm) with a tolerance of ±0.5 mm, ensuring manufacturing accuracy.

- Outer diameter: approximately 12 mm
- Sound port diameter: approximately 2.6 mm
- Total body length: approximately 9.5 mm
- Pin diameter: approximately 0.6 mm
- Pin spacing: approximately 4.0 mm

- Upper extension height: approximately 5.5 mm
- Total height: approximately 7.6 mm

The active buzzer contains an internal oscillator circuit. When a DC voltage is applied, it automatically generates a sound without the need for an external signal generator.

In the Smart Shopping Trolley system, the buzzer is activated by the ESP32 under the following conditions:

- Weight mismatch detected by the load cell
- Unauthorized product addition
- System error or warning condition

## 4.11 POWER SUPPLY (LI-ION BATTERY)

The system is powered using a rechargeable 18650 Li-ion battery. This provides portability and allows the trolley to operate without a fixed power source. A voltage regulator or boost converter is used to provide a stable 5V supply to the ESP32 and other components.



**Fig 4.22 18650 Li-ion Battery**

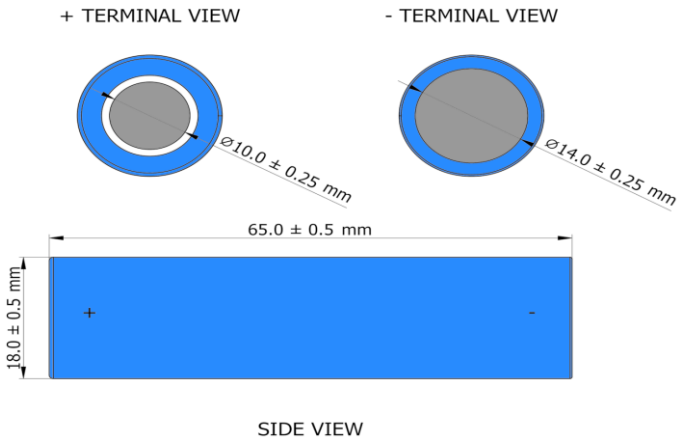



**Fig 4.23 2500mAh Battery Drawing & Dimensions**

| No. | Item | Specification | Unit | Notes |
|---|---|---|---|---|
| 2.1 | Normal Capacity<br>Minimum Capacity | 2500<br>2450 | mAh<br>mAh | 0.2CmA Discharge, 3.0V Cutoff |
| 2.2 | Nominal voltage | 3.70 | V | |
| 2.3 | Internal Impedance | ≤60 | mΩ | (with PCB) |
| 2.4 | Discharge Cut-off Voltage | 2.75 | V | |
| 2.5 | Charge Voltage | 4.20±0.05 | V | |
| 2.6 | End of Charge Current | 25 | mA | 0.01$C_5$, in CV mode |
| 2.7 | Standard Charge Current | 500 | mA | 0.2$C_5$, 0-45℃ |
| 2.8 | Maximum Charge Current | 1250 | mA | 0.5$C_5$, 20-45℃ |
| 2.9 | Standard Discharge Current | 500 | mA | 0.2$C_5$, 0-45℃ |
| 2.10 | Maximum Discharge Current | 2500 | mA | 1.0$C_5$, 20-45℃ |
| 2.11 | Maximum Instantaneous Discharge Current | 5000 | mA | 2.0$C_5$, 20-45℃ |
| 2.12 | Weight | ≈50 | g | |
| 2.13 | Operating Temperature Charge<br>Operating Temperature Discharge | 0 - +45<br>−10 - +60 | ◦C<br>◦C | |
| 2.14 | Storage Temperature (≤ 1 month)<br>Storage Temperature (≤ 6 months) | −5 - +45<br>0 - +45 | ◦C<br>◦C | |
| 2.15 | Standard Test Conditions | | | Temperature 23±2°C, Relative Humidity 65%±20% |

**Table 4.4 Battery Specifications**

The Lithium-Ion battery operates based on the movement of lithium ions between the anode and cathode during charging and discharging.

- During charging, lithium ions move from the cathode to the anode and store energy.
- During discharging, the ions move back to the cathode, releasing energy to power the circuit.

This electrochemical process provides a stable and efficient power supply to the system components.

## 4.12 BLOCK DIAGRAM

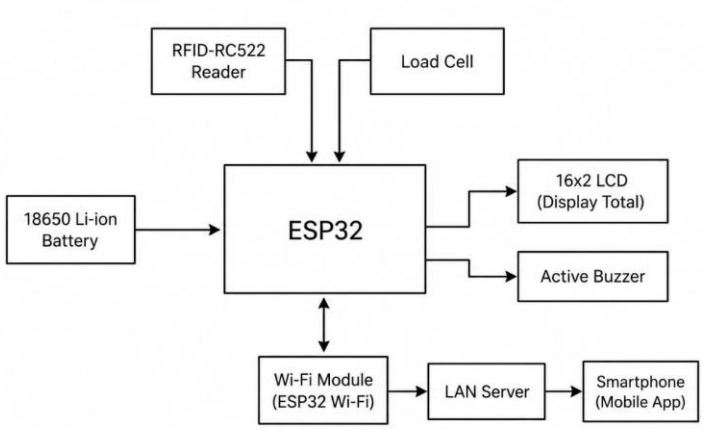



**Fig 4.24 block diagram of proposed system**

1. System Initialization (Power): The 18650 Li-Ion Battery provides continuous power to the central microcontroller ESP32 - waking up the brain of the trolley and activating all the connected peripherals.
2. Product Scanning (Data Input): When a shopper decides to buy an item, they tap the product's tag against the RFID-RC522 Reader. The reader extracts the unique product ID from the tag and transmits it to the microcontroller via SPI communication.

3. Weight Verification (Security Check): The shopper drops the item into the trolley. The Load Cell continuously monitors the physical weight of the basket's contents. The microcontroller cross-references the weight of the dropped item with the expected weight of the scanned RFID tag. This ensures that a shopper hasn't dropped an expensive, unscanned item into the cart.
4. Data Processing (The Brain): The ESP32 processes the incoming data. It matches the scanned RFID to its internal or fetched database, retrieves the price of the item, and adds it to the running calculation to formulate the new total bill.
5. Instant User Feedback (Local Output):
    a. Audio: The Active Buzzer emits a sharp beep, signal to the user that the scan was successful and the item was verified.
    b. Visual: The display module (16x2 LCD Display) refreshes instantly to show the shopper their updated cart status (Eg: "Total: 15.50").
6. Wireless Data Transmission (Data Transmit): The microcontroller leverages its built-in Wi-Fi capabilities to broadcast the updated cart data. It routes this information over a local LAN Server running inside the supermarket.

# CHAPTER 5

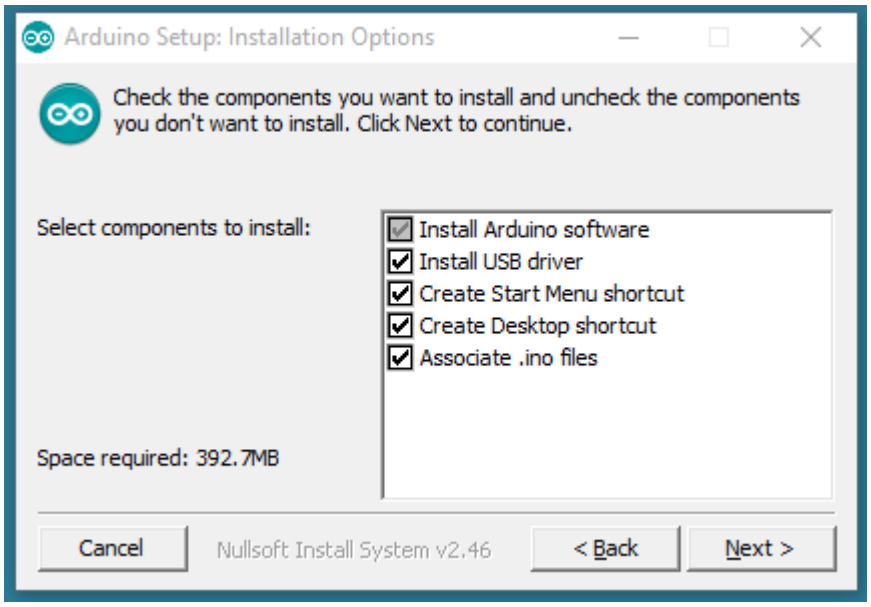
# SOFTWARE TOOLS

## 5.1 ARDUINO

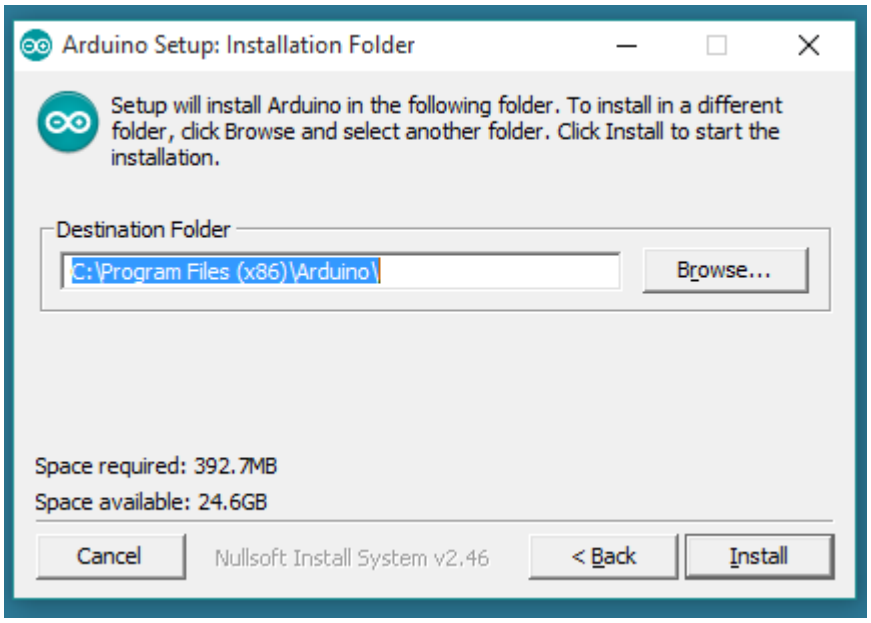
The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. It runs on Windows, Mac OS X, and Linux. The environment is written in Java and based on Processing and other open-source software. This software can be used with any Arduino board

## 5.2 INSTALLATION

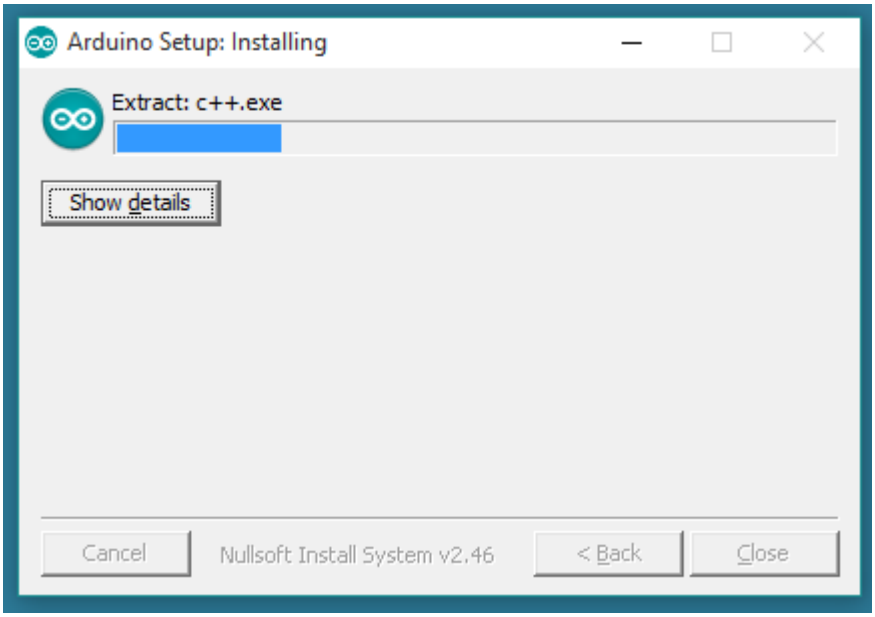
In this project Arduino 4.0is installed. We can choose between the Installer(.exe) and the Zip packages. If you use the first one that installs directly everything you need to use the Arduino Software (IDE), including the drivers. With the Zip package you need to install the drivers manually. The Zip file is also useful if you want to create a portable installation. When the download finishes, proceed with the installation and please allow the driver installation process when you get a warning from the operating system.




Choose the components to install




Choose the installation directory (suggested to keep the default one)




The process will extract and install all the required files to execute properly the Arduino Software (IDE)

## 5.3 LIBRARY INSTALLATION

### SoftwareSerial Library

The Arduino hardware has built-in support for serial communication on pins 0 and 1 (which also goes to the computer via the USB connection). The native serial support happens via a piece of hardware (built into the chip) called a UART. This hardware allows the Atmega chip to receive serial communication even while working on other tasks, as long as the room in the 64byte serial buffer.

The SoftwareSerial library has been developed to allow serial communication on other digital pins of the Arduino, using software to replicate the functionality (hence the name "SoftwareSerial"). It is possible to have multiple software serial ports with speeds up to 115200 bps. A parameter enables inverted signaling for devices which require that protocol.

To use this library, we have to write : **#include <SoftwareSerial.h>**

### 1. SPI Library

The SPI (Serial Peripheral Interface) library is used for communication between ESP32 and the RFID module.

**#include <SPI.h>**

Important Methods:

SPI.begin() //Initializes SPI communication.

SPI.beginTransaction() //Configures SPI settings before communication.

SPI.transfer(data) //Sends and receives data over SPI.

SPI.endTransaction() //Ends SPI communication.

**2. MFRC522 Library (RFID)**

This library is used to control the RFID RC522 module.

**#include <MFRC522.h>**

Important Methods:

mfrc522.PCD_Init() //Initializes the RFID reader.

mfrc522.PICC_IsNewCardPresent() //Checks if a new RFID card is detected.

mfrc522.PICC_ReadCardSerial() //Reads the unique ID (UID) of the RFID tag.

mfrc522.PICC_HaltA() //Stops communication with the card.

mfrc522.PCD_StopCrypto1() //Stops encryption after communication.

**3. Wire Library (I2C)**

The Wire library enables I2C communication for LCD.

**#include <Wire.h>**

Important Methods:

Wire.begin() //Initializes I2C communication.

Wire.beginTransmission(address) //Starts communication with a device.

Wire.write(data) //Sends data to the device.

Wire.endTransmission() //Ends communication.

**4. LiquidCrystal_I2C Library (LCD)**

This library is used to control the 16×2 LCD display.

**#include <LiquidCrystal_I2C.h>**

Important Methods:

lcd.init() //Initializes the LCD.

lcd.backlight() //Turns ON the LCD backlight.

lcd.setCursor(column, row) //Sets the cursor position.

lcd.print(data) //Displays text or values on the LCD.

lcd.clear() //Clears the display.

**5. HX711 Library (Load Cell)**

The HX711 library is used for weight measurement.

**#include "HX711.h"**

Important Methods:

scale.begin(DT, SCK) //Initializes the HX711 module.

scale.set_scale() //Sets calibration factor.

scale.tare() //Resets weight to zero.

scale.get_units() //Reads weight value.

scale.power_down() / power_up() //Controls power mode.

## 6. WiFi Library (ESP32)

The WiFi library is used for wireless communication.

**#include <WiFi.h>**

Important Methods:

WiFi.begin(ssid, password) //Connects ESP32 to WiFi network.

WiFi.status() //Checks connection status.

WiFi.localIP() //Returns assigned IP address.

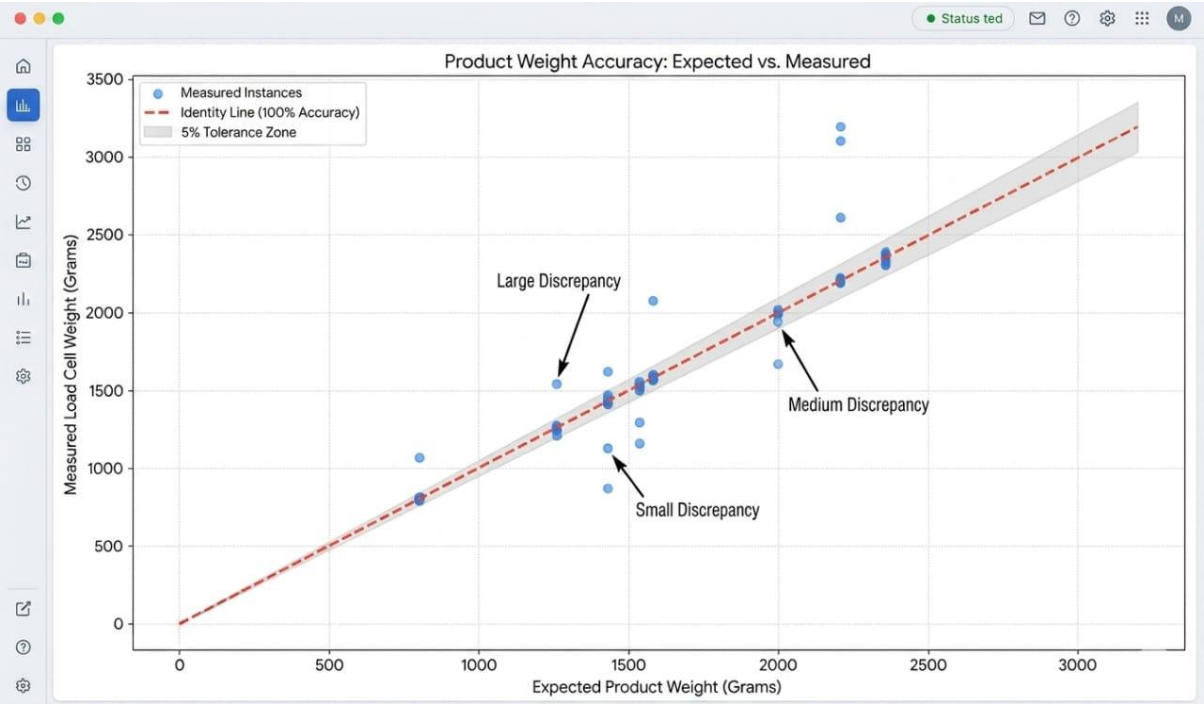
WiFi.disconnect() //Disconnects from network.

# CHAPTER 6

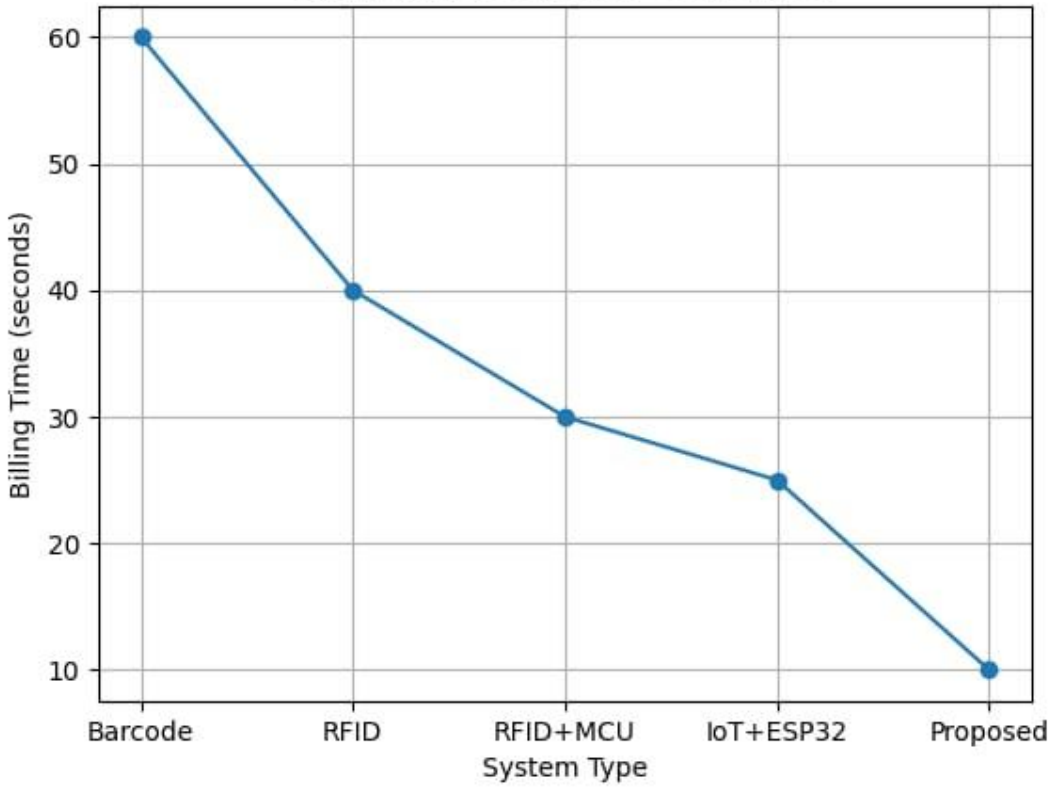
# RESULTS AND DISCUSSION

| TECHNOLOGY USED | METHOD/ APPROACH | RESULT/ OUTCOME | LIMITATIONS |
|---|---|---|---|
| Barcode System | Manual barcode scanning | Reduced manual errors | Checkout queue exists |
| RFID Technology | RFID-Based smart trolley | Faster scanning and billing | No verification system |
| Microcontroller + RFID | Automated billing with LCD | Real-time bill display | No anti-theft mechanism |
| IoT + ESP32 | Smart trolley with Wi-Fi | Real-time data monitoring | Increased system complexity |
| RFID + Load Cell | Weight verification system | Improved accuracy | Requires calibration |
| IoT + Mobile App | Digital payment integration | Enhanced user experience | High cost and complexity |
| RFID + Load Cell + ESP32 + Wi-Fi | Smart trolley with verification | Accuate billing, real-time display,theft prevention | Limited RFID range, minor weight varation |

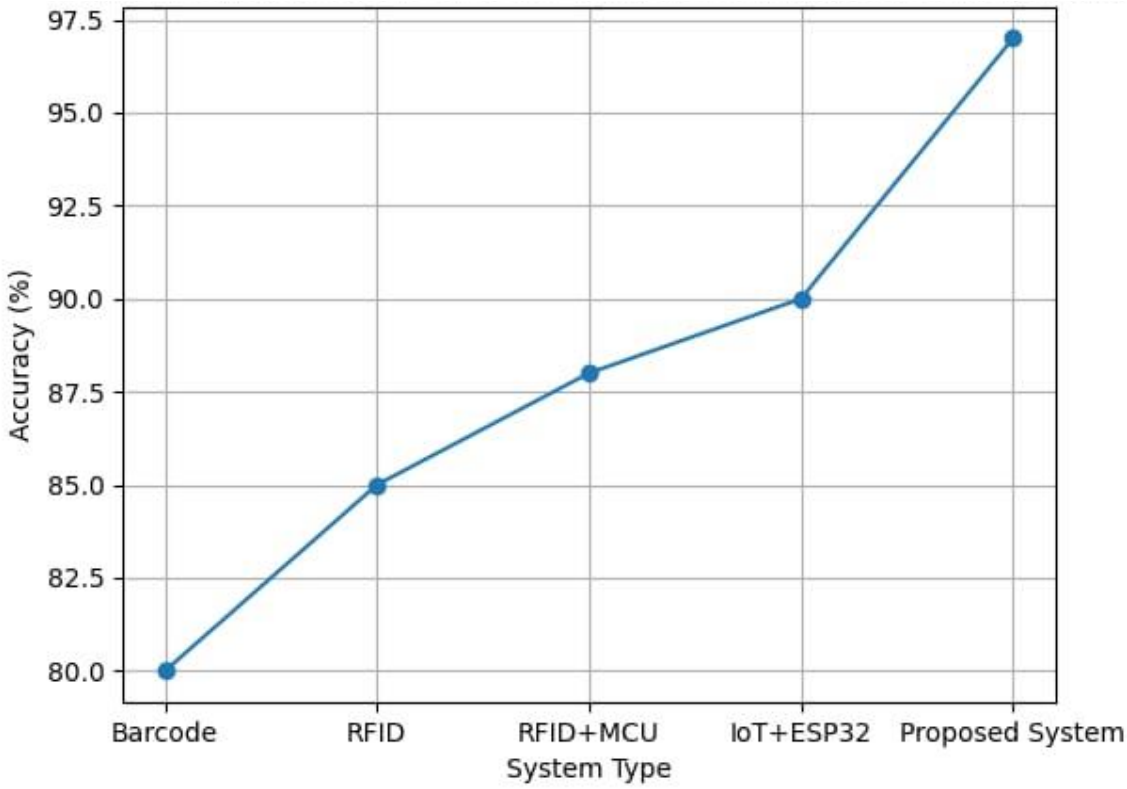
The above table presents a comparative analysis of smart shopping systems developed over the years. It shows a clear progression from traditional barcode-based systems to advanced RFID and IoT-based solutions.




# BILLING TIME REDUCTION COMPARISON



# COMPARSION OF PREVIOUS AND PROPOSED SYSTEM




The proposed system improves upon existing methods by integrating weight verification and real-time billing, resulting in enhanced accuracy, efficiency, and user convenience.This system was successfully implemented using an ESP32 microcontroller integrated with RFID, load cell, and LCD display, enabling automatic product identification and real-time billing. The system accurately scanned items, updated the total bill instantly, and verified product weight to prevent errors, with alerts generated in case of mismatches. It showed fast response and reliable performance, significantly reducing billing time and eliminating queues. Although minor limitations such as limited RFID range and slight weight variations were observed, the system proves to be an effective, user-friendly, and efficient solution for modern retail automation.


Smart Trolley
Connecting WiFi

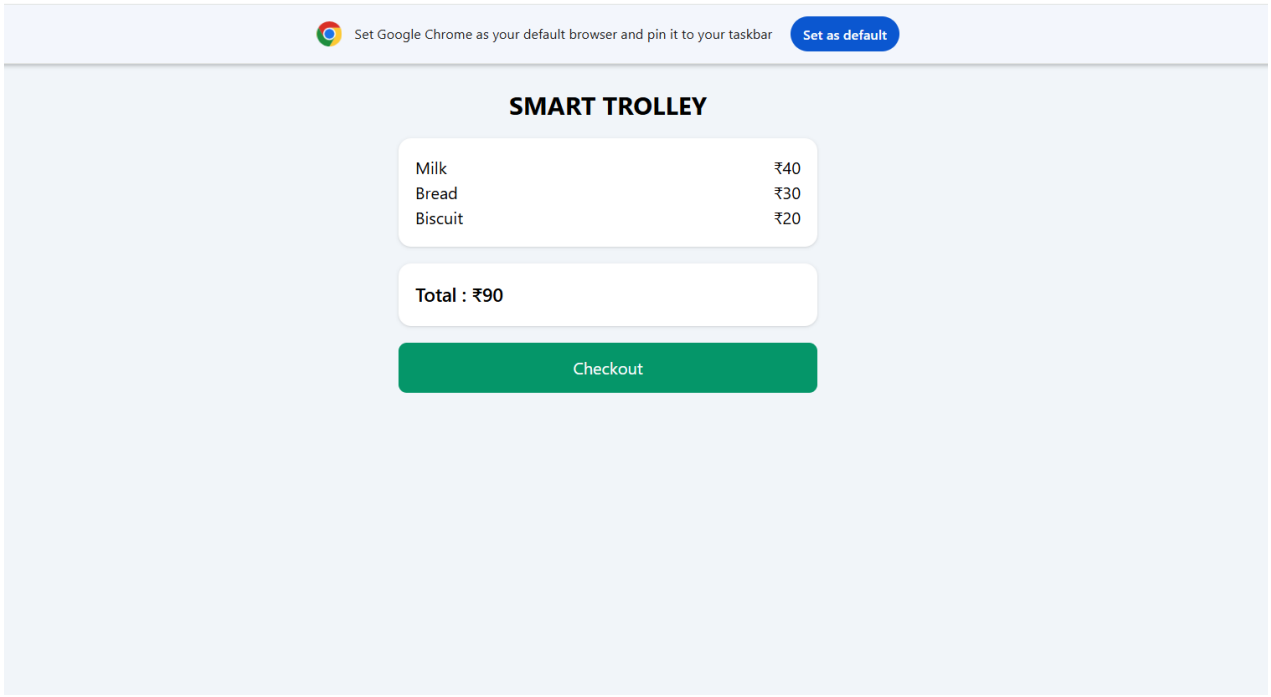

Scan Product
JHD 162A

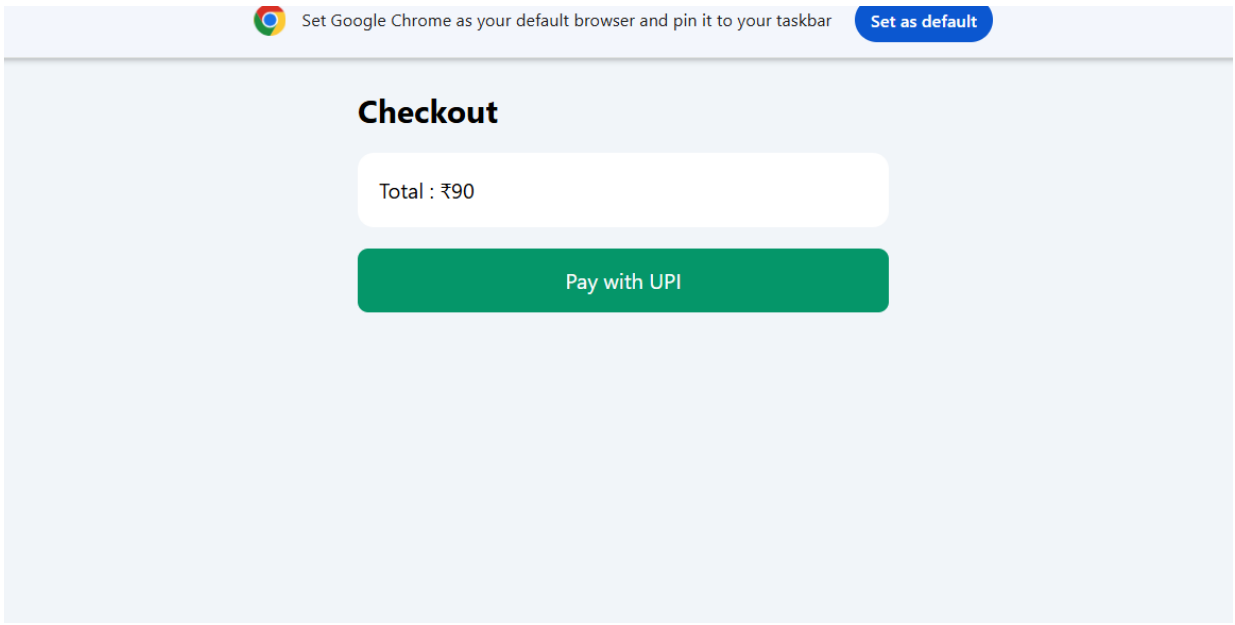

Bread Added
Total Rs:30

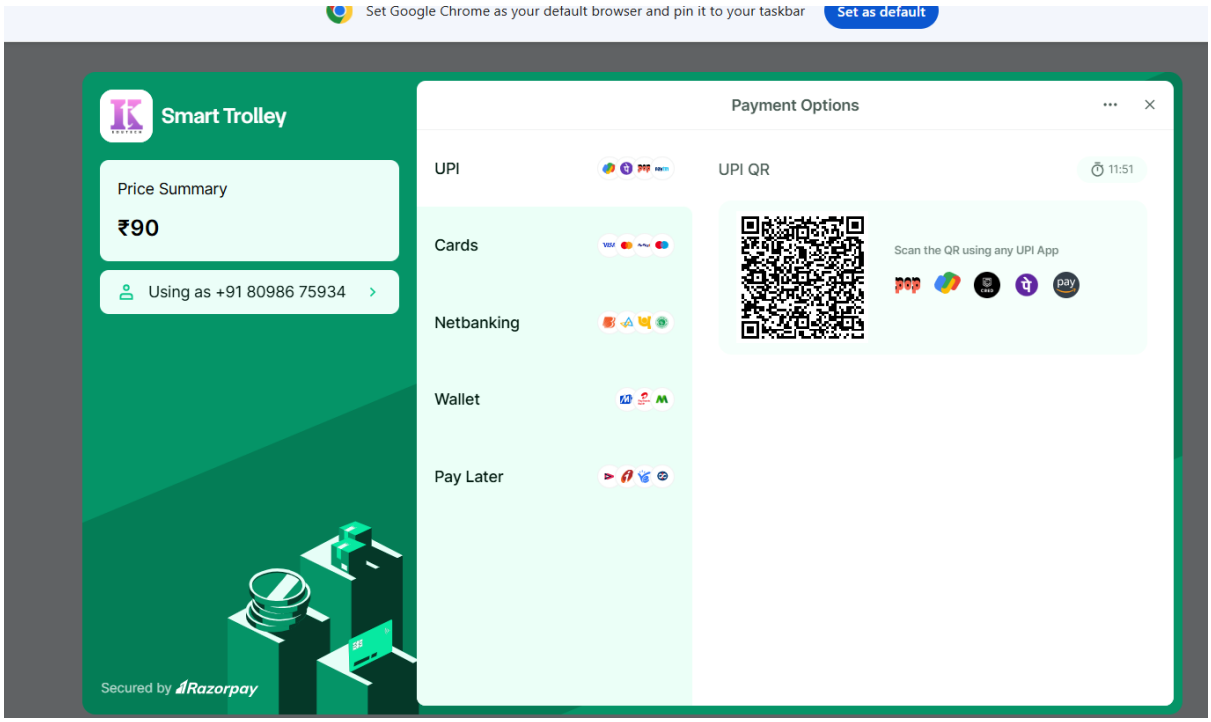

Biscuit Added
Total Rs:50


Milk Added
Total Rs:90


Biscuit Removed
Total Rs:70


Set Google Chrome as your default browser and pin it to your taskbar
Set as default
SMART TROLLEY
Milk ₹40
Bread ₹30
Biscuit ₹20
Total : ₹90
Checkout


Set Google Chrome as your default browser and pin it to your taskbar
Set as default
Checkout
Total : ₹90
Pay with UPI


Set Google Chrome as your default browser and pin it to your taskbar
Set as default
Smart Trolley
Price Summary
₹90
Using as +91 80986 75934
Secured by Razorpay
Payment Options
UPI
Cards
Netbanking
Wallet
Pay Later
UPI QR
11:51
Scan the QR using any UPI App

# CHAPTER 7

# CONCLUSION

This project is an innovative solution designed to improve the efficiency and convenience of the shopping experience in retail environments. By integrating technologies such as RFID for automatic product identification, a load cell for weight verification, and an ESP32 microcontroller for processing, the system enables real-time billing directly within the trolley. This eliminates the need for traditional checkout queues and significantly reduces billing time. The use of a 16×2 LCD display allows customers to view product details and the total amount instantly, enhancing transparency and helping users manage their purchases effectively. The inclusion of a weight verification mechanism improves system accuracy and helps prevent errors or unauthorized actions. Furthermore, the built-in Wi-Fi capability of the ESP32 provides opportunities for real-time monitoring and future integration with digital payment systems. Overall, of this project is a reliable, cost-effective, and user-friendly solution that enhances customer satisfaction and supports the development of smart retail environments.

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